# Survivorship Conditioning for Hazard Rates Driven at the Log-Derivative Level:
## A Two-Channel Model, a Two-Condition Viability Theorem, and the Critical Doob Limit

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November 2026 — Working Draft, rev. 7

**Abstract**

We study the survival theory of a hazard rate whose noise is placed one integration further out than is standard: rather than diffusing $h$ or $\log h$, we diffuse the negative logarithmic derivative $\mu = -\frac{d}{dt}\log h$, so that the integrated driver is a Kolmogorov diffusion and the governing persistence exponent moves from $\frac{1}{2}$ to $\frac{1}{4}$. The hazard carries two channels: an external background, muffled by the accumulated exponent, and an internal term $\Gamma e^{-\rho x}$ that the same exponent feeds or starves according to the sign of the response exponent $\rho$. The total hazard stays finite exactly when $\rho$ is positive and the driver's long-run behavior is favorable, so that no magnitude of accumulated capacity substitutes for the sign, and no event of bounded duration perfectly separates the survivorship-conditioned law from the unconditioned one. First, a trichotomy in $\rho$: for $\rho \leq 0$ the viability event $\{\Lambda_\infty < \infty\}$ is null for every parameter value and every driver, while for $\rho > 0$ viability factors as an indicator in $\rho$ times a driver-viability functional, two logically distinct necessary conditions. Second, a conjectural transfer: the two-channel model's survival asymptotics are conjectured to coincide with those of the bare integrated-driver model up to a shift of initial data and bounded constants, conditional on a soft-wall reduction stated as an explicit Ansatz, supported numerically and not proved. Third, the survivorship interpolation $\mathbb{P}(\cdot \mid T > t)$ is a monotone flow of upward tilts whose limit, in the critical regime, is the Doob $\mathsf{h}$-transform: locally absolutely continuous with respect to the unconditioned law on survivors and globally singular. Monte Carlo experiments support the $t^{-1/4}$ exponent and the finite-window kinematic factor. One candidate empirical signature follows, a survivor hazard decaying as $1/(4t)$ under criticality, which does not by itself distinguish the model from static frailty.

## 1 Introduction

Stochastic hazard rates are standard objects: the wrapper $S_t = \exp(-\int_0^t h)$ with $h$ a diffusion is the reduced-form framework of credit risk [1] and of stochastic mortality, where the force of mortality is routinely modeled by a diffusion, the mean-reverting Brownian Gompertz specification of Milevsky and Promislow [2] being a canonical instance. What distinguishes the present model is not the wrapper but *where the noise is placed*: standard practice diffuses $h$ or $\log h$, whereas

*Email: akb9408@rit.edu. This paper extracts, as a self-contained development in applied probability, the hazard model and survivorship theory of a larger framework; the companion developments carry the program built on them and are cited, not presupposed. Drafting collaboration: Anthropic Claude (Opus 4.x, Fable 5), under the author's direction.

we diffuse $\mu = -\frac{d}{dt}\log h$, one integration further out. The consequences are structural rather than cosmetic: $\log h$ becomes $C^1$, the state becomes a degenerate, 3/2-self-similar hypoelliptic diffusion, Kolmogorov's 1934 example [3, 4], and the governing persistence exponent moves from $\frac{1}{2}$ to $\frac{1}{4}$.

### 1.1 The model in three lines

*The hazard.* $h_t^{\mathrm{tot}} = B_t \varsigma_b(x_t) + \Gamma e^{-\rho x_t}$: an external background $B_t$ attenuated by a bounded logistic muffling factor $\varsigma_b$ that the accumulated exponent $x$ improves, plus an internal term that $x$ feeds or starves according to the sign of $\rho$.

*The driver.* $x_t = \int_0^t \mu_s\, ds$ with $\mu$ Brownian: the noise sits on $\mu = -\frac{d}{dt}\log h$. This placement is the paper's principal modeling claim.

*Departure.* $T$ is the first time the cumulative hazard exhausts an independent unit exponential (a Cox construction), and viability is the event $\{\Lambda_\infty < \infty\}$ that the total hazard ever accrued stays finite. Departure with probability one and infinite life expectancy are compatible, and both hold in the critical case.

*Position in the branch.* This paper is the root of a four-paper branch and imports nothing from the others: the theorem companion on target-conditioned records cites it for the survival specialization, the Route World companion for the conditioned law it hypothesizes governs the record, and the technoanatomy companion for the viability theorem and the ceiling; the hard-wall law $Q^{\mathrm{hard}}$ is proved here and the soft law $Q^{\mathrm{soft}}$ is conditional on Ansatz 4.3, and every downstream use says which it consumes.

*The claim.* The total hazard ever accrued stays finite exactly when the response exponent is positive and the driver's long-run behavior is favorable, so that no magnitude of accumulated capacity substitutes for the sign (Theorem 3.5), and no event of bounded duration on a surviving path perfectly separates the survivorship-conditioned law from the unconditioned one (Theorem 5.6). The rest of the paper establishes those two statements and the transfer that connects the two-channel model to the classical persistence theory they rest on.

### 1.2 Relation to the literature

Four bodies of work border this paper; what follows maps what is classical, what is adjacent, and what is claimed as new.

*The classical layer.* The state process is Kolmogorov's 1934 diffusion [3], the primordial hypoelliptic example [4]. Its persistence theory is a mature subject with a clean lineage, McKean [5], Goldman [6], Lachal [7], Sinai [8], Isozaki–Watanabe [9], Denisov–Wachtel [10], surveyed in [11] and, as the *random acceleration process*, possessing an extensive physics literature reviewed in [12]. The conditioned (never-hitting) process, our limit object in Section 5, is constructed by Groeneboom, Jongbloed and Wellner [13]. None of this is ours; we import it.

*The adjacent frameworks.* Our survivorship interpolation belongs to two established genres. First, Theorem 5.3 is a *penalization* statement in the sense of Roynette–Vallois–Yor [14]: a limit of Wiener-type measures reweighted by a time-dependent functional, identified as a martingale change of measure — here the weight is survival itself, and the polynomial tail places it in the clockless class. Second, the mode classification of Theorem 5.7 sits against the theory of quasi-stationary distributions and $Q$-processes surveyed in [15]: classical QSD theory is at home in the exponential-tail class, while the critical polynomial mode admits no proper QSD in fixed

coordinates, its Yaglom object being self-similar. We use both frameworks as positioning, not as inputs.

*The conceptual ancestor.* That selection on survival makes the observed ensemble diverge systematically from the unbiased one is the frailty phenomenon of Vaupel, Manton and Stallard [16] and Vaupel and Yashin [17]: when the frail are removed first, the population hazard falls below the individual hazard, producing mortality deceleration at the population level with no corresponding deceleration for any individual. We claim no priority over that insight, and regard this paper as exhibiting its *critical dynamical* analog: frailty is a static random variable fixed at birth and the analysis is a mixture computation, whereas here the randomness is a driver evolving in time and the object is a limit of path measures singular with respect to the unbiased law (Theorem 5.6).

*What is claimed as new.* To our knowledge, the following are new at least in assembly: the log-derivative placement of the noise, with its gauge analysis and the two-sided soft-wall analysis whose remaining gap is located exactly (Ansatz 4.2); the response trichotomy and the two-condition viability theorem (Theorem 3.5); and the survivorship interpolation read as a monotone flow with the tail index as its linear-response coefficient, classified into saturation, transformation and concentration (Theorem 5.7). Section 6 reports Monte Carlo support for the one non-rigorous reduction.

*Convention.* For positive functions, $f \sim g$ means $f/g \to 1$; $f \asymp g$ means $0 < \liminf f/g \leq \limsup f/g < \infty$. $\Phi$ is the standard normal distribution function. All processes live on a filtered space satisfying the usual conditions; $W$ is a standard one-dimensional Brownian motion. Claims are tagged: [P] proved here, [C] classical or cited, [A] conditional on Ansatz 4.2, [S] scaling-level, [O] open.

*Vocabulary.* Ten terms are the paper's own and are fixed here. *Departure*: the killing time $T$. *Viability*: the event $\{\Lambda_\infty < \infty\}$. *Credit*: the accumulated exponent $x_t = \int_0^t \mu$. *Response exponent*: $\rho$, the sign of which decides viability. *Muffling factor*: the bounded logistic $\varsigma_b(x)$ attenuating the background. *Driver-viability factor*: $V_{\mathrm{drv}}$, the second factor of Theorem 3.5, the driver's escape probability when it is transient and the indicator of a positive stationary mean when it is positive recurrent. *Hard wall / soft wall*: killing at first hitting of a level, against killing by accrued hazard. *Survivorship interpolation*: the family $\mathbb{P}(\cdot \mid T > t)$. *Bias field*: the drift the interpolation manufactures. *Modes*: saturation, transformation, concentration, the three limits of Theorem 5.7.

## 2 The model

*Why two channels.* The model's split of the hazard into an external channel and an internal one, with the internal channel's response to capacity carrying its own sign, records one observation that motivated it, the internal/external decomposition itself going back to [18]: once a civilization can destroy or redirect an asteroid on course to wipe out its planet, it has muffled an external hazard with a power that can equally be turned back on the biosphere as a weapon, so the same growth of capacity that answers the external channel opens the internal one, and maintaining the biosphere with stability, an internal rate low and robustly low, becomes a requirement that grows with capacity rather than one it discharges. The single exponent $\rho$ that carries that sign below is an idealization of a structural property the companion paper on technoanatomy decomposes further; the results here are stated for the scalar.

**Definition 2.1** (Driftless model)**.** [1] Fix $\sigma > 0$ and initial data $(\mu_0, D_0) \in \mathbb{R} \times \mathbb{R}$. Define

$$\mu_t = \mu_0 + \sigma W_t, \qquad D_t = D_0 + \int_0^t \mu_s \, ds, \qquad h_t = e^{-D_t}, \qquad \Lambda_t = \int_0^t h_u \, du. \tag{1}$$

The credit $D$ and the hazard are one coordinate: the initial hazard is $h_0 := e^{-D_0}$, and a multiplicative constant $q$ in the hazard is the translation $D_0 \mapsto D_0 - \log q$ of the initial credit, which is the content of Lemma 2.2. The hard wall of Section 4 sits at $D = 0$, where $e^{-D}$ passes through one. Let $E \sim \mathrm{Exp}(1)$ be independent of $W$ and $T := \inf\{t \geq 0 : \Lambda_t \geq E\}$, so that $\mathbb{P}(T > t \mid \mathcal{F}_\infty^W) = e^{-\Lambda_t}$. The *viability event* is $\mathsf{V} := \{\Lambda_\infty < \infty\}$; on $\mathsf{V}$ the conditional probability of immortality, $e^{-\Lambda_\infty}$, is strictly positive, and $\mathbb{P}(T = \infty) = \mathbb{E}[e^{-\Lambda_\infty} \mathbf{1}_{\mathsf{V}}] \leq \mathbb{P}(\mathsf{V})$.

The pair $(\mu, D)$ is the Kolmogorov diffusion, $\mathcal{L} = \frac{\sigma^2}{2}\partial_\mu^2 + \mu\,\partial_D$, Gaussian with covariance $\left(\begin{smallmatrix} \sigma^2 t & \sigma^2 t^2/2 \\ \sigma^2 t^2/2 & \sigma^2 t^3/3 \end{smallmatrix}\right)$ and hypoelliptic by Hörmander's condition at first order [3, 4]. Three structural facts are immediate and load bearing. There is no boundary in $h$: the logarithmic parametrization makes $h > 0$ automatic, and the whole problem concerns the growth of the integrated driver. The credit $D$ is nondecreasing exactly while $\mu \geq 0$: accumulated credit is spent only through a negative rate. And the initial hazard is exactly gauge:

**Lemma 2.2** (Gauge role of the initial credit; [P])**.** *Writing $\Lambda^{(D_0)}$ for the cumulative hazard from initial credit $D_0$, $\Lambda_t^{(D_0)} = e^{-D_0}\Lambda_t^{(0)}$ pathwise; hence $\mathsf{V}$ does not depend on $D_0$, while $\mathbb{P}(T = \infty)$ is continuous and nondecreasing in $D_0$, with $\lim_{D_0 \uparrow \infty} \mathbb{P}(T = \infty) = \mathbb{P}(\mathsf{V})$ and $\lim_{D_0 \downarrow -\infty} \mathbb{P}(T = \infty) = 0$. The initial credit, equivalently the initial hazard scale $h_0 = e^{-D_0}$, can never affect a dichotomy, only a value.*

*Remark* 2.3 (Nonparametric-Bayes reading). Definition 2.1 places an integrated Wiener prior on $-\log h$ — the classical cubic smoothing-spline prior of Wahba [19] — so the results below may be read as the survival asymptotics that canonical smoothness prior implies. A stronger reading is available: writing $\mathcal{D}[f] := -\frac{d}{dt}\log f$, one has $\mathcal{D}[\bar{F}] = h$ and $\mathcal{D}[h] = \mu$, so the driver is the second iterate of the operation that defines the hazard, and the model occupies a rung of a ladder rather than a member of a family. Noise at level one ($\log h$ Brownian) gives persistence exponent $\frac{1}{2}$; at level two, the present model, $\frac{1}{4}$; at level three the credit is twice-integrated Brownian motion and the exponent is open [11, 20]. Level two is the last rung that is both nontrivial and solvable.

**Definition 2.4** (Two-channel model)**.** [2] Fix $\sigma, \epsilon > 0$, $n \in \mathbb{N}$, $\Gamma > 0$, $\rho \in \mathbb{R}$, and offset $b > 0$. With $\mu$, $x$ as in (1) (writing $x$ for the integrated driver), let $B_t = \sum_{i=1}^n e^{\epsilon W_t^{(i)}}$ for Brownian

[1] The explicit reason I chose to use the logarithmic derivative of the hazard rate for the Wiener noise in the first place was for better consistency with the Lindy effect, which makes the system want to "hold" its intrinsic parameters for a time. That is, in that my original statement of the Golden Point already emphasizes the local capacity in a civilization for self-preservation, (with this capacity intuitively being related to the negative logarithmic derivative of the hazard rate, which after all remains a positive constant in a biosphere whose hazard rate decays exponentially), then applying the Lindy effect to this capacity itself has you expect to see each next metaphorical second typically undergo a similar proportional shift in the hazard rate as did the previous one, (since intuitively you would expect the central tendency to continue this progression nonconfrontationally rather than veer off of it suddenly). Hence, oscillations should probably be in this capacity itself as opposed to in the hazard rate directly.

[2] The two-channel form and the viability criterion $\Lambda_\infty < \infty$ have a dated germ in the author's document of August 2022 [18], where the criterion appears as $e^{a/b} < \infty$ under a two-channel "Outlook"; the present formulation replaces the germ and is cited for priority only.

motions $W^{(i)}$ independent of one another and of the driver $W$, an assumption the infinite-set lemma below requires when the set is generated by the driver, for independent Brownian motions $W^{(i)}$, and

$$h_t^{\mathrm{tot}} = \underbrace{B_t\, \varsigma_b(x_t)}_{\text{external}} + \underbrace{\Gamma\, e^{-\rho x_t}}_{\text{internal}}, \qquad \varsigma_b(x) := \frac{1+e^{-b}}{1+e^{x-b}}, \tag{2}$$

normalized so that $\varsigma_b(0) = 1$ exactly. Survival, $T$, and $\mathsf{V}$ are as before with $h^{\mathrm{tot}}$ in place of $h$.

**Proposition 2.5** (Channel dominance; boundedness; [P]). *(i) For every $\rho > 0$, $\Gamma e^{-\rho x}/(B\varsigma_b(x)) \to \infty$ as $x \to -\infty$, so the background drops out of the total at large negative $x$ for every positive $\rho$. (ii) As $x \to +\infty$ the external channel dominates for all large $x$ if and only if $\rho > 1$; for $\rho < 1$ the two cross at $x_\times = (k+b)/(1-\rho)$, $k := \log(B/\Gamma)$. (iii) $0 < \varsigma_b(x) < 1 + e^{-b}$ for all $x$: the external channel can be reduced without bound but amplified by at most $1 + e^{-b} \to 1$. The internal channel is unbounded above.*

*Remark* 2.6 (The causal asymmetry). Item (iii) enforces structurally a constraint a multiplicative form $Be^{-x}$ would violate: a failure of the system's own muffling cannot amplify the external world, while nothing bounds self-inflicted hazard. In operator terms, at fixed background, $\mathcal{D}[h^{\mathrm{int}}] = \rho\,\mu$ exactly, while $\mathcal{D}[h^{\mathrm{ext}}] = \varsigma(x-b)\,\mu$ with logistic weight: the internal channel preserves the defining operator relation with constant gain $\rho$, and the external channel must be deformed because boundedness is being purchased. This identifies $\rho$ intrinsically as the internal channel's operator gain, $\rho = -\partial_x \log h^{\mathrm{int}}$: the rate at which accumulated exponent converts into change in self-inflicted hazard. The companion program names $x$ and $\rho$ *technostrength* and *technoanatomy*, in the downstream mechanism companion; here they are the muffling exponent and the response exponent.

## 3 The response trichotomy and the two-condition theorem

We now let $\rho$ range over $\mathbb{R}$.

**Proposition 3.1** (Trichotomy; [P]). *Write $h(x) = B\varsigma_b(x) + \Gamma e^{-\rho x}$ with $B, \Gamma > 0$.*

(i) *If $\rho > 0$, $h$ is strictly decreasing with $h(x) \to 0$ as $x \to +\infty$.*

(ii) *If $\rho = 0$, $h$ is strictly decreasing with $h(x) \to \Gamma > 0$: protective against a floor.*

(iii) *If $\rho < 0$, $h \to B(1+e^{-b})$ as $x \to -\infty$ and $h \to \infty$ as $x \to +\infty$, so that $\inf_x h(x) > 0$ for frozen $B > 0$, which is all the fatality theorem consumes. In the regime $|\rho| > 1$ there is a unique interior critical point, a minimum, whose location is the large-$x$ asymptote of the exact critical-point equation $Ky^{r-1}(1+y)^2 = 1$ below, obtained from $\varsigma_b(x) \sim \mathrm{const}\cdot e^{-x}$ (writing $A := Be^b$, $|\rho| = -\rho$):*

$$x_* \;\sim\; \frac{\log\big(A/(|\rho|\Gamma)\big)}{1+|\rho|} \qquad (x_* - b \to +\infty), \qquad h_* \;\sim\; A^{\frac{|\rho|}{1+|\rho|}}\,\Gamma^{\frac{1}{1+|\rho|}}\Big(|\rho|^{\frac{1}{1+|\rho|}} + |\rho|^{-\frac{|\rho|}{1+|\rho|}}\Big). \tag{3}$$

*For $|\rho| \le 1$ the number of interior critical points depends on the parameters and may be zero: with $y = e^{x-b}$ and $r = |\rho|$ they solve $Ky^{r-1}(1+y)^2 = 1$ for a positive constant $K$, whose left side has logarithmic derivative $r - 1 + 2y/(1+y)$, increasing throughout when $r \ge 1$ and decreasing then increasing when $r < 1$, so that for $r < 1$ there are zero, one tangential, or two solutions, and for $r = 1$ none when $K \ge 1$. Nothing below depends on the count.*

**Lemma 3.2** (Infinite sets defeat the driftless background; [P]). *Let $W$ be a standard Brownian motion, $\epsilon > 0$, and let $S \subset [0,\infty)$ be a measurable set, possibly random but independent of $W$, with $\mathrm{Leb}(S) = \infty$ almost surely. Then $\int_S e^{\epsilon W_u}\, du = \infty$ almost surely; consequently the same holds with $B_u = \sum_{i=1}^n e^{\epsilon W_u^{(i)}} \geq e^{\epsilon W_u^{(1)}}$ in place of $e^{\epsilon W_u}$.*

*Proof.* Conditioning on $S$, it suffices to treat deterministic $S$ of infinite measure. For each $T$, finiteness of $\int_{S\cap[T,\infty)} e^{\epsilon W_u} du$ is unchanged by the multiplicative factor $e^{\epsilon W_T}$ — it is decided by $\int_{S\cap[T,\infty)} e^{\epsilon(W_u - W_T)} du$ — and the integral over $S \cap [0,T]$ is a.s. finite, so the event $\{\int_S e^{\epsilon W_u} du = \infty\}$ lies in $\sigma(W_u - W_T : u \geq T)$ for every $T$, hence in the tail $\sigma$-field of the increments, and has probability 0 or 1 by Kolmogorov's zero–one law. Were the probability 0, then $\int_S e^{\epsilon W_u} du < \infty$ a.s., and applying the same to $-W$, also $\int_S e^{-\epsilon W_u} du < \infty$ a.s.; adding, $2\,\mathrm{Leb}(S) \leq \int_S (e^{\epsilon W_u} + e^{-\epsilon W_u}) du < \infty$, contradicting $\mathrm{Leb}(S) = \infty$. □

**Proposition 3.3** (Recurrent drivers; [P]). *Suppose the driver of §2 is positive recurrent with stationary law $\pi$ and $\int |\mu|\, \pi(d\mu) < \infty$; write $\bar{m} := \int \mu\, \pi(d\mu)$.*

(a) *If $\bar{m} > 0$ then $D_t/t \to \bar{m}$ a.s., $\Lambda_\infty < \infty$ a.s., and $\mathbb{P}(\mathsf{V}) = 1$.*

(b) *If $\bar{m} < 0$ then $\Lambda_\infty = \infty$ a.s. and $T < \infty$ a.s.*

(c) *If $\bar{m} = 0$ and the cycle integral defined in the proof is nondegenerate (automatic for nondegenerate diffusions), then $\Lambda_\infty = \infty$ a.s.:* the critical case falls on the mortal side.

*For null-recurrent drivers no general statement is offered **[O]**; the Brownian driver, the fundamental null-recurrent case, is settled unconditionally by the driftless analysis of §4.*

*Proof.* (a), (b): the ergodic theorem for positive recurrent one-dimensional diffusions gives $D_t/t \to \bar{m}$ a.s. (see [21, Ch. X]); if $\bar{m} > 0$ then eventually $D_t \geq D_0 + \frac{\bar{m}}{2} t$ and $\Lambda_\infty < \infty$; if $\bar{m} < 0$ the integrand $e^{-(D_t - D_0)} \to \infty$.

(c) Fix interior levels $a < b$ and regenerate at up-crossings: let $R_0 := \inf\{t : \mu_t = b\}$ and inductively $S_k := \inf\{t > R_{k-1} : \mu_t = a\}$, $R_k := \inf\{t > S_k : \mu_t = b\}$. By the strong Markov property the cycles over $[R_{k-1}, R_k)$ are i.i.d.; positive recurrence gives $\mathbb{E}[L] < \infty$ for the cycle length $L := R_1 - R_0$, and the cycle occupation identity yields, for the cycle integral $\xi_k := \int_{R_{k-1}}^{R_k} \mu_s\, ds$,

$$\mathbb{E}[\xi_1] = \mathbb{E}[L] \int \mu\, d\pi = 0, \qquad \mathbb{E}|\xi_1| \leq \mathbb{E}\Big[\int_{\text{cycle}} |\mu_s| ds\Big] = \mathbb{E}[L] \int |\mu|\, d\pi < \infty.$$

Thus $S_n := D_{R_n} - D_{R_0} = \sum_{k\leq n} \xi_k$ is a nondegenerate mean-zero i.i.d. walk, so $\liminf_n S_n = -\infty$ a.s. by the classical oscillation dichotomy for random walks (see, e.g., [22]); in particular the events $B_n := \{D_{R_n} \leq -1\}$ occur infinitely often a.s. Choose $\delta > 0$ and $M < \infty$ with

$$q := \mathbb{P}_b\Big(L \geq \delta,\ \sup_{0\leq u\leq \delta} \Big|\int_0^u \mu_s\, ds\Big| \leq M\Big) > 0,$$

possible since $L > 0$ a.s. and the supremum is finite a.s. Let $G_n$ be the corresponding event for cycle $n{+}1$; by the strong Markov property $\mathbb{P}(G_n \mid \mathcal{F}_{R_n}) = q$, so $\sum_n \mathbb{P}(B_n \cap G_n \mid \mathcal{F}_{R_n}) = q \sum_n \mathbf{1}_{B_n} = \infty$ a.s., and Lévy's extension of the Borel–Cantelli lemma (see, e.g., [22]) gives $B_n \cap G_n$ infinitely often a.s. On $B_n \cap G_n$, for $u \in [0,\delta]$ one has $D_{R_n+u} \leq -1 + M$, hence

$$\int_{R_n}^{R_n+\delta} e^{-D_s}\, ds \ \geq\ \delta\, e^{\,1-M-D_0} \ >\ 0,$$

and these windows are disjoint. Summing over infinitely many of them, $\Lambda_\infty = \infty$ a.s. □

**Theorem 3.4** (Non-positive response is fatal, and fast; [P])**.** *Let $\rho \le 0$. Then: (i) $\Lambda_\infty = \infty$ almost surely, for the random background and regardless of the driver — pathwise in the driver, almost surely in the background. (ii) If the background is frozen, $B_t \equiv B > 0$, then $h_* = \inf_x h(x) > 0$ and $\mathbb{P}(T > t) \le e^{-h_* t}$, whence $\mathbb{E}[T] < \infty$. For the random background no positive floor exists ($\inf_t B_t = 0$ a.s.), so the exponential rate is frozen-background (annealed) only, and whether an unconditional rate holds is open [O]. (iii) Whether any power-law structure survives in the random-background case is open [O]: almost-sure death does not exclude polynomial survival tails, and the exclusion is proved only in the frozen-background setting of (ii).*

*Proof.* (i) If $x_t \to +\infty$ the internal term diverges pointwise (for $\rho < 0$; for $\rho = 0$ it is the constant $\Gamma$). If $x_t \to -\infty$ the external term tends to $B_t(1 + e^{-b})$ and $\int_0^\infty B_t\, dt = \infty$ a.s. by Lemma 3.2 with $S = [0, \infty)$. On any other driver path, either $\mathrm{Leb}\{t : x_t \ge 0\} = \infty$, where the trough floors the hazard at $\Gamma e^{-\rho x_t} \ge \Gamma$ deterministically, or $\mathrm{Leb}\{t : x_t \le 0\} = \infty$, where the crest with $S = \{t : x_t \le 0\}$ routes through Lemma 3.2. (ii) $\Lambda_t \ge h_* t$ and $\mathbb{P}(T > t) = \mathbb{E}[e^{-\Lambda_t}] \le e^{-h_* t}$. □

**Theorem 3.5** (Two conditions for viability; [P])**.** *Let the driver be a regular diffusion on $\mathbb{R}$ with scale function $s$, and $\rho \in \mathbb{R}$. Define the* driver-viability functional

$$V_{\mathrm{drv}}(\mu_0) = \begin{cases} \mathbb{P}_{\mu_0}(\mu_t \to +\infty) = \dfrac{s(\mu_0) - s(-\infty)}{s(+\infty) - s(-\infty)}, & \textit{transient driver,} \\ 1, & \textit{positive recurrent, } \bar{m} > 0, \\ 0, & \textit{positive recurrent, } \bar{m} \le 0, \\ 0, & \textit{Brownian driver,} \\ \textit{open [O]}, & \textit{general null recurrence,} \end{cases} \tag{4}$$

*with $\bar{m} = \int \mu \, d\pi$ the stationary mean where it exists, the transient formula read with its boundary conventions ($s(+\infty) = \infty$ and $s(-\infty)$ finite giving 0; $s$ bounded above with $s(-\infty) = -\infty$ giving 1). Then*

$$\mathbb{P}\big(\Lambda_\infty < \infty\big) = \underbrace{\mathbf{1}\{\rho > 0\}}_{\textit{response}} \cdot \underbrace{V_{\mathrm{drv}}(\mu_0)}_{\textit{driver}}. \tag{5}$$

*The two factors are logically distinct necessary conditions, not independent events: the first is a sign, the second is silent on it. In the positive recurrent case the driver does not escape; its time integral acquires positive linear drift, which is why the functional is stated piecewise rather than as a single scale-function ratio.*

*Proof.* On $\{\mu_t \to +\infty\}$: eventually $\mu_t \ge 1$, so $x_t \ge x_{t_0} + (t - t_0)$ and, for $\rho > 0$, $h(x_t) \lesssim e^{-\min(1,\rho)(t - t_0)}$, whence $\Lambda_\infty < \infty$. For $\rho \le 0$, Theorem 3.4(i) makes the first factor necessary. On $\{\mu_t \to -\infty\}$, $x_t \to -\infty$ and $h \to B(1 + e^{-b}) > 0$, so $\Lambda_\infty = \infty$; on the recurrent event the outcome is Proposition 3.3 — $\bar{m} > 0$ gives $D_t/t \to \bar{m}$ and $\Lambda_\infty < \infty$, $\bar{m} \le 0$ (nondegenerate) gives $\Lambda_\infty = \infty$ — with the Brownian driver settled by the driftless analysis and general null recurrence left open, which is the statement's scope clause. The hitting probabilities are the classical scale-function formula [23, 21]. Throughout, the background $B$ is random and is never fixed: for positive linear growth the argument uses the subexponential growth of the Brownian background, and for adverse growth it uses the infinite-set lemma above. □

**Corollary 3.6** (The solvable case; [P]). *For the self-reinforcing driver $d\mu_t = \nu\mu_t\,dt + \sigma\,dW_t$, $\nu > 0$ — the member of the driver family whose drift is proportional to its state — $s'(v) = e^{-\nu v^2/\sigma^2}$ is integrable at both ends and*

$$\mathbb{P}\big(\Lambda_\infty < \infty\big) = \mathbf{1}\{\rho > 0\} \cdot \Phi\Big(\frac{\mu_0\sqrt{2\nu}}{\sigma}\Big)\,: \tag{6}$$

*a sharp threshold in the response exponent multiplied by a smooth Gaussian sigmoid of width $\sigma/\sqrt{2\nu}$ in the initial rate. With $\rho < 0$, feedback accelerates failure: $\mu_t \to +\infty$ drives $h \to \infty$ superexponentially, so a system with adverse response and feedback is strictly worse off than one with adverse response and none.*

*Remark* 3.7 (Fluctuating response; [S]). Theorem 3.5 is stated for constant $\rho$. It applies to a process $\rho_t$ exactly when $\rho_t$ is eventually constant, its response factor then reading $\mathbf{1}\{\rho_\infty > 0\}$ for the eventual value $\rho_\infty$, which is the form in which the technoanatomy companion consumes it; it says nothing about a response that keeps fluctuating, and the rest of this remark says what can be said in that case. If $\rho = \rho_t$ is a process, an adverse excursion of duration $\delta$ at exponent level $x$ contributes $\asymp \delta\,\Gamma e^{|\rho|x}$ to $\Lambda$, so survivability requires $\delta \lesssim \Gamma^{-1}e^{-|\rho|x}$: the affordable duration of adverse response decays exponentially in the accumulated exponent. Recurrence of $\rho_t$ alone does not force $\Lambda_\infty = \infty$: the internal term explodes only where adverse anatomy coincides with positive credit, which is the overlap condition of Corollary 3.8. Viability does not, however, require $\rho_t$ eventually positive: a viable driver with $x_t/t \to 1$ and anatomy negative only on intervals $[n, n + e^{-3n}]$ has negative anatomy recurring arbitrarily late with summable hazard contributions, and the general question is the integrability of $e^{-\rho_t x_t}$; whether (5) admits a product decomposition over anatomy and driver is a separate conjecture, not asserted here. Supplying dynamics for $\rho_t$ is the model's principal open direction [O].

**Corollary 3.8** (Adverse anatomy overlapping positive credit is fatal; [P]). *If there exist $a, b > 0$ with $\mathrm{Leb}\{t : \rho_t \le -a,\ x_t \ge b\} = \infty$ almost surely, then $\Lambda_\infty = \infty$ almost surely, since on that set the internal term satisfies $\Gamma e^{-\rho_t x_t} \ge \Gamma e^{ab}$. A negative excursion of $\rho$ that systematically coincides with $x_t \le 0$ costs nothing, so recurrence of $\rho$ is not by itself fatal; what fluctuating anatomy implies in general, when the excursions of $\rho$ and of $x$ are not coupled by hypothesis, is open [O].*

**Proposition 3.9** (Perturbation in $\Gamma$, and the threshold; fixed-time expansion [P], long-time threshold [O]). *Writing $Q_{\mathrm{ext}}$ for the external-only survivorship measure,*

$$\mathbb{P}(T > t) = \mathbb{P}_{\mathrm{ext}}(T > t)\Big[1 - \Gamma\,\mathbb{E}_{Q_{\mathrm{ext}}}\Big[\int_0^t e^{-\rho x_s}ds\Big] + O(\Gamma^2)\Big], \tag{7}$$

*the fixed-$t$ expansion being a Taylor–Feynman–Kac identity. Under $Q_{\mathrm{ext}}$ the credit grows as $x_s \asymp \sigma s^{3/2}$ on the scaling picture, which suggests that the first-order coefficient converges as $t \to \infty$ if and only if $\rho > 0$; the suggestion is not a proof, since $\mathbb{E}_Q[e^{-\rho x_s}]$ is controlled by the mass the conditioned process keeps near low credit and not by its typical scale, and the long-time threshold is entered as open. Were it established, the expansion's radius would collapse exactly at the trichotomy's boundary, the signature of a genuine transition.*

**Theorem 3.10** (Adverse credit occupies half of logarithmic time in the critical regime; [P]). *For the driftless driver, almost surely*

$$\lim_{S\to\infty} \frac{1}{S}\,\mathrm{Leb}\big\{s \in [0, S] : D_{e^s} - D_0 < 0\big\} = \frac{1}{2}, \tag{8}$$

*so the set of times at which the credit sits below its initial value has logarithmic density exactly one half, for every initial datum. The statement is confined to the critical recurrent regime and fails for any driver whose credit is transient.*

*Proof.* $Z_s := (D_{e^s} - D_0)/(\sigma e^{3s/2})$ is a stationary, centered, Gaussian, ergodic process, by the Lamperti transform of the integrated Brownian motion under its 3/2-self-similarity, so Birkhoff's ergodic theorem gives logarithmic density $\mathbb{P}(Z_0 < 0) = \frac{1}{2}$ by symmetry. □

For general $\mu_0$ the rescaled increment is $(D_{e^s} - D_0)/(\sigma e^{3s/2}) = (\mu_0/\sigma)\, e^{-s/2} + e^{-3s/2} \int_0^{e^s} W_u \, du$, so the displayed process is exactly stationary and centered only when $\mu_0 = 0$; the proof applies ergodicity to the stationary component and then uses that the deterministic shift $(\mu_0/\sigma)e^{-s/2}$ vanishes, which leaves the limiting sign occupation unchanged, so the conclusion holds for every $\mu_0$.
The theorem is exported for the technoanatomy companion, which reads it as the statement that adverse strength is a generic condition in the critical regime and not evidence of adverse anatomy.

## 4 Persistence and the transfer

**Theorem 4.1** (Persistence of the hard-wall functional; [C])**.** *Let $\tau := \inf\{t \geq 0 : D_t \leq 0\}$ for the driftless model started at $\mu_0 \geq 0$, $D_0 > 0$. Then $\mathbb{P}_{(\mu,d)}(\tau > t) \sim c\, \mathsf{h}(\mu, d)\, t^{-1/4}$ as $t \to \infty$, where $\mathsf{h}$ is the persistence-harmonic function of the killed Kolmogorov semigroup, with boundary behavior $\mathsf{h}(\mu, 0^+) \propto \mu^{1/2}$ and $\mathsf{h}(0^+, d) \propto d^{1/6}$, and $\mathbb{E}_x[\mathsf{h}(X_s)\mathbf{1}_{\{\tau > s\}}] = \mathsf{h}(x)$. The exponent is Sinai's [8]; the refined form and the harmonic function are McKean, Goldman, Lachal, Isozaki–Watanabe, and Groeneboom–Jongbloed–Wellner [5, 6, 7, 9, 13]; universality is Denisov–Wachtel [10], surveyed in [11, 12].*

The soft-killing functional replaces the hard wall by the killing rate $e^{-D}$, exponentially negligible for $D \gg 0$ and overwhelming for $D \ll 0$: an absorbing wall of thickness $O(1)$ in $D$ at the level $D = 0$ where the hard wall $\tau = \inf\{t : D_t \leq 0\}$ sits, negligible against excursions of size $t^{3/2}$. Killing approximates absorption when the credit crosses downward through zero, and the sign convention is fixed here once so that it cannot drift. The reduction is stated as two hypotheses of increasing strength, because the exponent and the conditioned law require different things: the tail asymptotics rest on the weaker and the $Q$-process on the stronger.

**Ansatz 4.2** (A: soft killing preserves the exponent)**.** For every admissible initial datum there exist $0 < c \leq C < \infty$ and $t_0$ with $c\,\mathbb{P}(\tau > t) \leq \mathbb{P}(T > t) \leq C\,\mathbb{P}(\tau > t)$ for $t \geq t_0$, after an $O(1)$ adjustment of the initial credit. Equivalently $\log \mathbb{P}(T > t) = \log \mathbb{P}(\tau > t) + O(1)$.

**Ansatz 4.3** (B: soft survival has a ratio asymptotic)**.** Writing $K_t f(x) = \mathbb{E}_x[e^{-\int_0^t q(X_u)du} f(X_t)]$ for the Feynman–Kac semigroup with $q(\mu, D) = e^{-D}$, there is a positive function $g$ on the state space and a constant $c_{\text{soft}}$ with $K_t 1(x) \sim c_{\text{soft}}\, g(x)\, t^{-1/4}$ locally uniformly in $x$; formally $g$ is the zero-energy positive solution of $(\mathcal{L} - q)g = 0$ with the growth conditions the soft wall imposes.

Ansatz 4.2 does not imply Ansatz 4.3: comparability fixes $\log \mathbb{P}_x(T > t)$ to $O(1)$ but leaves the hidden constant free to depend on $x$, so that $\mathbb{P}_x(T > t) \sim c(x)\mathsf{h}(x)t^{-1/4}$ with $c$ non-constant is compatible with A and incompatible with the conclusion that the soft conditioned law is the hard-wall $\mathsf{h}$-transform. The question the two hypotheses leave open, which the numerical

agenda of Section 6 targets, is not whether $g = C\,\mathsf{h}(\cdot + \delta)$ for some effective wall translation $\delta$, which exact proportionality the operators obstruct, since $\mathcal{L}\mathsf{h} = 0$ makes a translated $\mathsf{h}$ still $\mathcal{L}$-harmonic while positive $g$ satisfies $\mathcal{L}g = qg > 0$, but whether $g$ approaches a translated $\mathsf{h}$ in an appropriately specified asymptotic or scaling limit, which is the question the numerics target.

One half is rigorous modulo a cited input: on the event $\{D_u \geq 2\log\frac{2+u}{2}\ \forall u \leq t\}$ the cumulative hazard is bounded, and a logarithmic moving boundary does not change the persistence exponent of an integrated process [11, 10]; the upper half reduces to the plausible, unproved statement that persistence with a bounded occupation tolerance has the same exponent as strict persistence. The nearest standing literature cuts against assuming this lightly: persistence with *partial survival*, each crossing killing with probability $p$, carries exponents varying continuously with $p$ [24], for the random acceleration process among others [25], so softening a wall can shift an exponent. What distinguishes the present softening is its structure: killing is rate-based and confined to a boundary layer of width $O(1)$ in the credit against excursions of scale $t^{3/2}$, a vanishing fraction of the range, where partial survival taxes every crossing at fixed price at every scale. Everything tagged [A] depends on exactly this gap, and Section 6 tests it directly; the measured constancy of the soft-to-hard ratio is evidence the boundary-layer case sits in the exponent-preserving class the moving-boundary results delimit from the rigorous side.

**Proposition 4.4** (Transfer; scaling statement [S], prefactor conjectural)**.** *In the two-channel model with $\rho > 1$ the persistence exponent $1/4$ is expected to survive the two-channel perturbation, because the $t^{3/2}$ scale of the integrated driver dominates the $t^{1/2}$ scale of the background's fluctuations, so that under Ansatz 4.2 $\log\mathbb{P}(T > t) = -\frac{1}{4}\log t + O(1)$. A heuristic derivation, not reproduced here and not relied on, suggests the more specific form*

$$\mathbb{P}(T > t) \asymp \mathsf{h}\big(\mu_0 - \tfrac{\epsilon^2}{2},\, D_0 - \log n - b\big) \cdot e^{-\Theta(n(b/\sigma)^{2/3})} \cdot t^{-1/4}, \tag{9}$$

*every difference between the two-channel model and the driftless one appearing as a shift of initial data or a bounded multiplicative constant; that form is entered here as a conjecture, its shifts and prefactor not established by the hypotheses stated in this paper, and the numerics below test the bare soft wall rather than the full two-channel process. Where the exponent holds, $\mathbb{E}[T] = \infty$ while $T < \infty$ a.s., residual life is asymptotically Pareto($\frac{1}{4}$), and quantiles grow as the fourth power of rarity.*

## 5 Conditioning on survival

**Definition 5.1** (Survivorship interpolation; bias field)**.** $\mathbb{P}^{(t)} := \mathbb{P}(\cdot \mid T > t)$, likewise $\mathbb{P}_\tau^{(t)}$ for the hard wall; for $0 \leq s < t$ the bias field on $[0, s]$ is

$$\beta_t(s) := \frac{d\mathbb{P}^{(t)}}{d\mathbb{P}}\Big|_{\mathcal{F}_s} = \mathbf{1}_{\{T>s\}}\,\frac{\mathbb{P}_{X_s}(T > t - s)}{\mathbb{P}_{X_0}(T > t)}, \qquad X_s = (\mu_s, D_s). \tag{10}$$

**Lemma 5.2** (Survivorship tilts upward; [P])**.** *Coupling two initial data componentwise with the same $W$ shows both $\mathbb{P}_x(T > t)$ and $\mathbb{P}_x(\tau > t)$ nondecreasing in $(\mu_0, D_0)$; hence every bias field is, on the survival event, a nondecreasing function of the state: the interpolation is a monotone flow of upward tilts in rate and credit.*

**Theorem 5.3** (The critical limit is the pure Doob transform; [C], soft version [A])**.** *Under the refined asymptotics of Theorem 4.1, for every $s \geq 0$ and bounded $\mathcal{F}_s$-measurable $\Psi$, $\mathbb{E}[\Psi \mid \tau >$*

$t] \to \mathbb{E}_Q[\Psi]$ *as* $t \to \infty$, *where* $Q$ *is the time-homogeneous Markov law with*

$$\frac{dQ}{d\mathbb{P}}\Big|_{\mathcal{F}_s} = \mathbf{1}_{\{\tau>s\}} \frac{\mathsf{h}(X_s)}{\mathsf{h}(X_0)} : \tag{11}$$

*the pure Doob* $\mathsf{h}$*-transform at eigenvalue one. Under Ansatz 4.3, and conditionally on it, the soft survivorship law* $\mathbb{P}(\cdot \mid T > t)$ *converges to the Feynman–Kac Doob transform*

$$\frac{dQ^{\mathrm{soft}}}{d\mathbb{P}}\Big|_{\mathcal{F}_s} = e^{-\int_0^s q(X_u)du} \frac{g(X_s)}{g(X_0)}, \tag{12}$$

*which does not coincide with* (11) *even when* $g$ *is proportional to* $\mathsf{h}$ *after a credit translation: the hard density vanishes on paths crossing the wall while the soft density remains positive on finite paths with finite accumulated killing and positive* $g$*, so the two conditioned laws have different supports and are distinct laws. Ansatz 4.2 alone yields neither the limit nor its existence.*

*Proof.* With $f_t := \mathbf{1}_{\{\tau>s\}}\mathbb{P}_{X_s}(\tau > t-s)/\mathbb{P}_{X_0}(\tau > t)$, the Markov property gives $\mathbb{E}[f_t] = 1$; factorizing, $f_t = \mathbf{1}_{\{\tau>s\}} \frac{(t-s)^{1/4}\mathbb{P}_{X_s}(\tau>t-s)}{t^{1/4}\mathbb{P}_{X_0}(\tau>t)}(1-s/t)^{-1/4} \to f := \mathbf{1}_{\{\tau>s\}}\mathsf{h}(X_s)/\mathsf{h}(X_0)$ a.s.; the martingale property gives $\mathbb{E}[f] = 1$, so $f_t \to f$ in $L^1$ by Scheffé, and $\mathbb{E}[\Psi f_t] \to \mathbb{E}_Q[\Psi]$. □

*Remark* 5.4 (Why survivorship has no clock here; [P]). For regularly varying tails the kinematic factor $\mathbb{P}_x(\tau > t-s)/\mathbb{P}_x(\tau > t) \to 1$ for fixed $s$: the age of the conditioning drops out and the limit is an un-tilted $h$-transform. For exponential tails the same factor converges to $e^{\theta s}$ and survivorship acquires a clock. In the penalization taxonomy [14] the survival weight lands in the clockless class precisely because its expectation is regularly varying. Quantitatively

$$\beta_t(s) = \mathbf{1}_{\{\tau>s\}} \frac{\mathsf{h}(X_s)}{\mathsf{h}(X_0)} \Big(1-\frac{s}{t}\Big)^{-1/4}(1+o(1)), \qquad \Big(1-\frac{s}{t}\Big)^{-1/4} = 1 + \frac{s}{4t} + O\Big(\frac{s^2}{t^2}\Big) : \tag{13}$$

the persistence exponent is the linear-response coefficient of survivorship bias.

**Theorem 5.5** (Survivorship drift; promotion to transience; [P] given [C] inputs)**.** *Under* $Q$ *the state solves* $d\mu_t = \sigma^2 \partial_\mu \log \mathsf{h}(\mu_t, D_t)\, dt + \sigma\, dW_t^Q$, $dD_t = \mu_t\, dt$*: conditioning adds drift only in the noisy coordinate, the drift is everywhere nonnegative, equals* $\sigma^2/2\mu$ *near zero credit (the Bessel-2 form, the halving of the classical exponent under one integration), and vanishes as* $\kappa := \sigma^2 D/\mu^3 \to \infty$*. Under* $\mathbb{P}$ *the credit is oscillatory,* $\liminf D_s = -\infty$ *and* $\limsup D_s = +\infty$ *(recurrence classifications belong to the Markov pair, and* $D$ *alone is not Markov); under* $Q$ *it is transient,* $D_s \to \infty$ *a.s.: the Pólya* 2D → 3D *transition with no literal change of dimension anywhere — survivorship alters the recurrence class of the ensemble, not the state dynamics.*

**Theorem 5.6** (Locally ours, globally elsewhere; [P/C])**.** $Q(\tau = \infty) = 1$*;* $\mathsf{h}(X_s) \to \infty$ *and* $D_s \to \infty$ $Q$*-a.s. [13];* $\liminf_s D_s = -\infty$ $\mathbb{P}$*-a.s. Hence* $Q|_{\mathcal{F}_s} \ll \mathbb{P}|_{\mathcal{F}_s}$ *for every finite* $s$*, with the explicit density* (11) *and mutual absolute continuity on the survival event, while* $Q \perp \mathbb{P}$ *on* $\mathcal{F}_\infty$*: the survivor's law is locally equivalent to the unconditioned one on survivors and globally disjoint from it, the exit from the measure class completed only at* $t = \infty$*. Consequently no event of bounded duration perfectly separates the conditioned law from the unconditioned one: none has probability one under one law and zero under the other on their common survival support. Mutually absolutely continuous laws can still be far apart statistically, so this is a statement about certification and not about discrimination.*

**Theorem 5.7** (Three canonical survivorship regimes; [P/C/S]). *Three tail classes of the driver produce three regimes of the interpolation, canonical rather than exhaustive: stretched-exponential and other subexponential tails, and regularly varying tails of exponent other than 1/4, fall under the same categories at the level of the survival weight's behavior in the penalization taxonomy but are not treated here.* (I) Saturation *[P] (favorable:* $\mathbb{P}(T=\infty)>0$*):* $\|\mathbb{P}^{(t)}-\mathbb{P}(\cdot \mid T=\infty)\|_{\mathrm{TV}}\to 0$ *with every bias bounded by* $1/\mathbb{P}(T=\infty)$*; no new measure class.* (II) Transformation *[C/A] (regularly varying tails that also admit a statewise ratio asymptotic,* $\mathbb{P}_x(T>t)\sim g(x)L(t)t^{-\alpha}$*; regular variation of the scalar tail alone does not determine the state-dependent ratio): the clockless Doob or Feynman–Kac transform by* $g$*, locally absolutely continuous, globally singular, with its kinematic factor; the hard-wall case* $\alpha=1/4$*,* $g=\mathsf{h}$*, is the proved instance (Theorem 5.3) and the soft case is conditional on Ansatz 4.3.* (III) Concentration *[S] (exponential tails,* $\mathbb{P}(T>t)\sim e^{-\theta t}$*): on fixed windows the eigen-tilted transform, the quasi-stationary regime with a clock; for the linearly adverse driver* $d\mu=\nu\,dt+\sigma\,dW$ *with* $\nu<0$ *the clock is* $\theta=3\nu^2/8\sigma^2$ *and, at macroscopic scale, the conditioned path law collapses onto the deterministic optimal profile* $\mu_{tv}/t\to\frac{|\nu|}{4}v(2-3v)$*, a law of large numbers for the rare, a large-deviation computation for the linearly adverse driver that is stated here at scaling level and not derived. Modes (II) and (III) share* $\mathbb{P}(\mathsf{V})=0$ *and are separated by the interpolation — by whether the survivor's measure class is approached polynomially, without a clock, or exponentially, with one.*

**Proposition 5.8** (Survivorship conditioning selects any linearly growing hazard gap; [P]). *Let the histories split into two classes,* $C_1$ *of prior weight* $\alpha\in(0,1)$ *and* $C_0$ *of weight* $1-\alpha$*, whose hazards are realized on one probability space, with cumulative hazards* $\Lambda^{(1)},\Lambda^{(0)}$ *and class survival functions* $S_j(t)=\mathbb{E}\big[e^{-\Lambda_t^{(j)}}\big]$ *as in the Cox construction, and suppose that for some* $\Delta>0$ *and* $K<\infty$

$$\Lambda_t^{(0)}-\Lambda_t^{(1)}\ \geq\ \Delta t-K\qquad \text{for all } t\geq 0,\ \text{almost surely.} \tag{14}$$

*Then the odds of* $C_0$ *against* $C_1$ *under the interpolation satisfy*

$$O(\eta):=\frac{\mathbb{P}(C_0\mid T>\eta)}{\mathbb{P}(C_1\mid T>\eta)}\ \leq\ \frac{1-\alpha}{\alpha}\,e^{K-\Delta\eta}, \tag{15}$$

*so that* $O(\eta)<\beta$ *for every* $\eta>\eta_*:=\Delta^{-1}\big[\log\frac{1-\alpha}{\alpha}+\log\frac{1}{\beta}+K\big]$*, which is finite for every* $\alpha>0$*, and* $\mathbb{P}(C_1\mid T>\eta)\to 1$ *as* $\eta\to\infty$*. The bound holds with equality for constant hazards differing by* $\Delta$ *with* $K=0$*. In particular (14) holds with* $\Delta=\delta h_{\min}$ *when* $\Lambda^{(1)}\leq(1-\delta)\Lambda^{(0)}+K$ *for some* $\delta\in(0,1]$ *and the hazard of* $C_0$ *has a floor* $h_{\min}>0$*.*

*Proof.* By (14), $e^{-\Lambda_t^{(0)}}\leq e^{K-\Delta t}\,e^{-\Lambda_t^{(1)}}$ on every path; taking expectations, $S_0(t)\leq e^{K-\Delta t}S_1(t)$, and by Bayes $O(\eta)=(1-\alpha)S_0(\eta)/\big(\alpha S_1(\eta)\big)$. In the fractional case $\Lambda_t^{(0)}-\Lambda_t^{(1)}\geq\delta\Lambda_t^{(0)}-K\geq\delta h_{\min}t-K$. □

*Remark* 5.9 (What the hypothesis needs; the two-channel instance [P]). Rarity enters only as $\log(1/\alpha)$: a class of prior weight $10^{-12}$ whose hazard sits a hundredth of the floor below the other's has odds below $10^{-6}$ once $\eta$ exceeds about $4.1\times10^3/h_{\min}$. Both hypotheses are needed. Without a floor the higher class could itself be viable, $\Lambda_\infty^{(0)}<\infty$ with positive probability, and a fractional drop would leave the odds at a positive limit, as in mode (I) of Theorem 5.7. And the comparison must hold along paths, the lower hazard carried below the other with its fluctuations, and not merely in mean: a hazard that fluctuates more widely can outsurvive a lower and steadier one, since $\mathbb{E}[e^{-\Lambda}]\geq e^{-\mathbb{E}[\Lambda]}$ with the gap in the survivor's favor, so a lower

mean hazard alone selects nothing. In the two-channel model of Definition 2.4 with $\rho > 0$ the hypothesis has a direct instance. Along a common background path, credit held at or above $x_* > 0$ gives $h^{\mathrm{tot}} \le (1-\delta)(B_t + \Gamma)$ with $1-\delta = \max\{\varsigma_b(x_*), e^{-\rho x_*}\}$, while credit at or below zero leaves $h^{\mathrm{tot}} \ge B_t + \Gamma \ge \Gamma$, since $\varsigma_b \ge 1$ and $e^{-\rho x} \ge 1$ there. A class that holds the one and a class confined to the other therefore satisfy the fractional hypothesis with floor $\Gamma$, the background's independence of the driver supplying the coupling, and any positive held credit is a positive drop. Holding credit indefinitely carries positive weight only where the driver's long-run behavior is favorable, the second factor of Theorem 3.5.

*Remark* 5.10 (The conditioned law read as stated content). Each horizon event $\{T > t\}$ has positive probability, and the limit of Theorem 5.3 is the law that has absorbed every one of them. Read not as a limit of laws but as the content of an information state, it states survival through every horizon at once, every other attribute blank: in set notation $\bigcap_t \{T > t\} = \{T = \infty\}$, null in the critical regime, which is why it is reached through the interpolation and never conditioned on as an event. The larger program in which these results are interpreted, whose vocabulary this paper otherwise does not use, calls that content the Golden Aethus at its primitive, and takes as its Golden Axiom that it is stated in one's own information state: an exposed postulate, carried in the downstream companions at the grade of an open hypothesis [O] and nowhere asserted or used here. Theorem 5.6 says what it would cost, since no event of bounded duration separates the conditioned law from the unconditioned one with certainty, and no finite record could therefore certify the statedness; Proposition 5.8 says what it would buy, since under the conditioning any class whose hazard sits linearly below another's takes all of the weight.

## 6 Numerical experiments

The one load-bearing heuristic, Ansatz 4.2, admits a direct test, and three falsifiers are fixed before any result is read: a fitted soft-wall slope departing robustly from $-1/4$; a dependence of that slope on the initial hazard $h_0$; and instability of the conditioned rescaled credit across late windows. Any one firing would count against the Ansatz. All experiments use the exact pair simulation: for a step $\Delta$, the increments $(\xi, \eta) = (W_{t+\Delta} - W_t, \int_t^{t+\Delta}(W_s - W_t)ds)$ are centered Gaussian with covariance $\left(\begin{smallmatrix} \Delta & \Delta^2/2 \\ \Delta^2/2 & \Delta^3/3 \end{smallmatrix}\right)$, so the update $\mu \mathrel{+}= \sigma\xi$, $D \mathrel{+}= \mu\Delta + \sigma\eta$ reproduces the law at grid points with zero discretization error; the only approximations are the trapezoidal quadrature of $\Lambda$ and discrete monitoring of the wall. Parameters: $\sigma = 1$, $\mu_0 = 0$, with the soft-wall curves labeled by the initial hazard $h_0 = e^{-D_0}$ (so $h_0 \in \{0.3, 1, 3, 30\}$ is $D_0 \in \{1.20, 0, -1.10, -3.40\}$). The soft-wall curves are not separate runs: they come from the one run at $D_0 = 1$ with the hazard rescaled by $h_0 e^{D_0}$, which is exact for the soft wall by Lemma 2.2, a multiplicative constant in the hazard being a translation of the initial credit; $N = 1.5 \times 10^5$ paths; a uniform grid of step 0.025 to $t = 40$ (the zone where the $\Lambda$-mass accrues), geometric thereafter at ratio 1.01 to $t = 3 \times 10^4$; slopes fitted by least squares on $\log_{10} \mathbb{P}$ against $\log_{10} t$ over $t \in [10^2, 3 \times 10^4]$, with path-bootstrap confidence intervals (150 resamples). A control run at half the step sizes (0.0125, ratio 1.005) reproduces the estimates below to within their intervals, excluding discretization bias.

Three findings deserve emphasis. First, the soft-killing functional carries the $t^{-1/4}$ exponent at $h_0 \in \{0.3, 1, 3\}$, with no dependence on $h_0$ beyond sampling error (Figure 1b), and the argument rests on these three. The $h_0 = 30$ curve is not resolved at this sample size: its survival probability is of order $10^{-11}$, which $1.5 \times 10^5$ paths cannot estimate, so the estimate rests on a handful of paths and its fitted slope moves with the window, $-0.298$ over $t \ge 10^2$, a value

| quantity | measured | predicted | status |
|---|---|---|---|
| hard-wall slope, $\log \mathbb{P}(\tau > t)$ | $-0.2499$ $[-0.254, -0.243]$ | $-1/4$ (theorem) | validates code |
| soft-wall slope, $h_0 = 0.3$ | $-0.2487$ | $-1/4$ [A] | supports |
| soft-wall slope, $h_0 = 1$ | $-0.2480$ $[-0.255, -0.242]$ | $-1/4$ [A] | supports |
| soft-wall slope, $h_0 = 3$ | $-0.2483$ | $-1/4$ [A] | supports |
| soft-wall slope, $h_0 = 30$ | not resolved at this $N$ ($\mathbb{P} \sim 10^{-11}$) | $-1/4$ by Lemma 2.2 | not tested |
| ratio $\mathbb{P}(T > t)/\mathbb{P}(\tau > t)$, $h_0 = 1$, $t \geq 10^2$ | $\in [0.1768, 0.1788]$ | bounded [A] | supports, sharply |
| kinematic coefficient (grid-exact) | $0.2874 \pm 0.0133$ | 0.2859 | confirms (13) |
| KS distance, $D_t/t^{3/2}$ at $t = 2 \times 10^3$ vs $3 \times 10^4$ | 0.010 | $\to 0$ (Yaglom) | consistent |

Table 1: Monte Carlo results ($N = 1.5 \times 10^5$; a second seed reproduces the slope estimates to $\pm 0.004$). The kinematic test evaluates $(\mathbb{P}(\tau > t-s)/\mathbb{P}(\tau > t) - 1)(t/s)$ at the grid spacing $s/t = 0.197$, whose exact power-law value is $((1-s/t)^{-1/4} - 1)(t/s) = 0.2859$; the measurement matches the full factor, not merely its linearization $1/4$.

inside the initial transient, and $-0.24$ or $-0.12$ over later windows. It neither supports nor counts against the Ansatz; there the exponent is fixed by Lemma 2.2, which is exact, and a rare-event estimator (importance sampling on the driver) is the computation that would resolve it. Second, and this is the sharpest support for Ansatz 4.2, the ratio $\mathbb{P}(T > t)/\mathbb{P}(\tau > t)$ is not merely bounded over the fitted two decades but constant to within about one per cent, drifting slowly from 0.1768 to 0.1788 (Figure 1c), consistent with convergence to a limit constant $\approx 0.178$ at these initial data — a stronger behavior than the Ansatz asserts. Third, the survivorship drift's qualitative structure is confirmed: among paths surviving to the horizon, the conditional increment $\mathbb{E}[\Delta\mu]$ regressed on $1/\mu$ over a window $\Delta t = 50$ at $t = 2 \times 10^3$ gives a positive coefficient of the predicted order at small $\kappa$ ($51 \pm 27$ against the instantaneous-drift value 25; the finite window biases upward, as conditioned near-wall paths must gain rate over it) and a coefficient indistinguishable from zero at $\kappa > 2$ ($0.1 \pm 0.4$), matching $G(\kappa) \to 0$. The remaining sharp test, the prefactor constant in (9) and the deep-tail behavior at $t/t_* \gtrsim 10^{12}$ via multilevel splitting or harmonic-function importance sampling, is left to a dedicated computation. The experiments test Ansatz 4.2, the exponent; they do not test Ansatz 4.3. The computation that would is fixed here: estimate $R_t(x, y) = \mathbb{P}_x(T > t)/\mathbb{P}_y(T > t)$ over a grid of initial states $(\mu, D)$ at increasing $t$, and ask whether it converges to $\mathsf{h}(x)/\mathsf{h}(y)$, or to $\mathsf{h}(x + \delta)/\mathsf{h}(y + \delta)$ for a fixed translation, or to a ratio of some other positive $g$; with harmonic-function importance sampling the deep tail is reachable. *None of the three falsifiers fired*: no fitted slope departs robustly from $-1/4$, no dependence of the slope on $h_0$ appears among the resolved values, and the conditioned rescaled credit is stable.

## 7 Discussion

The claim stated at the outset stands as proved: viability holds exactly when the response exponent is positive and the driver's long-run behavior is favorable, no magnitude of accumulated capacity substituting for the sign, and no event of bounded duration on a surviving path perfectly separates the conditioned law from the unconditioned one. One empirical signature follows. Under criticality the population hazard of survivors is $-\frac{d}{dt} \log \mathbb{P}(T > t) \sim \frac{1}{4t}$: mortality deceleration not to a plateau but to zero, at a universal rate set by the persistence exponent.

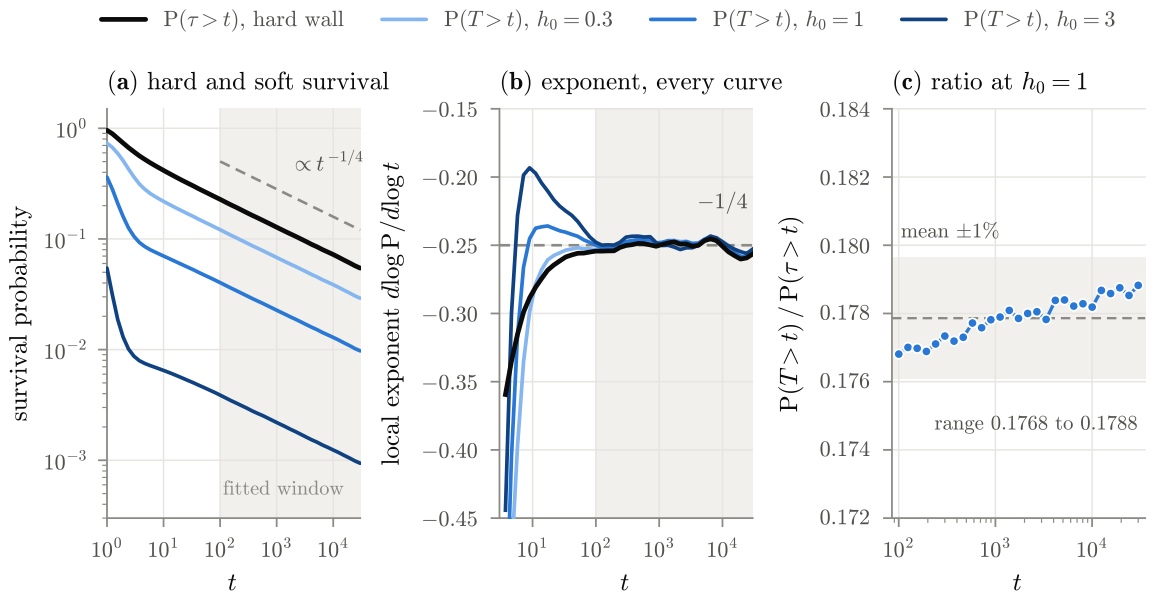



Figure 1: (a) Hard-wall survival and soft-wall survival at $h_0 \in \{0.3, 1, 3\}$, with the $t^{-1/4}$ reference and the fitted window $t \geq 10^2$ shaded. (b) The local exponent $d \log \mathbb{P} / d \log t$ of each curve, converging to $-1/4$. (c) The soft-to-hard ratio at $h_0 = 1$ over the fitted window: constant to within about 1% across two decades. The $h_0 = 30$ curve, not resolved at this sample size, is not drawn.

The reliability-theory literature on late-life mortality deceleration [26] documents plateaus as the fingerprint of static frailty selection; a hazard decaying as $c/t$ with $c$ universal would be the corresponding fingerprint of the dynamical mechanism, and distinguishing the two in longevity data is a concrete empirical question this model makes precise.

The theory assembled here is the critical dynamical analog of the frailty phenomenon [16, 17]: where frailty computes how an observed hazard curve bends when a static susceptibility is selected on, the present interpolation identifies the limiting law of the survivor's entire trajectory, together with the drift conditioning manufactures, and locates the observed-versus-unbiased divergence in the measure class itself. The two-condition theorem gives the model's structural moral in one line: an index of accumulated capacity tracks the driver-viability factor and is silent on the sign that multiplies it, so no such index can distinguish the ascent that arrives from the ascent that is a fall — and with $\rho < 0$, feedback strictly accelerates the fall. The interpretation of these results within a larger program, where $x$ and $\rho$ are read as a biosphere's capability and its disposition to turn capability on itself, and the conditioning as survivorship over deep time, is developed in the downstream companions of this branch, on which nothing above depends and none of which is cited here.

Five problems are open, and they are the paper's every [O] tag gathered in one place. Supplying dynamics for $\rho_t$ (Remark 3.7) would convert the trichotomy's sign test into a two-transience condition and is the model's principal open direction. Closing Ansatz 4.2, persistence with bounded occupation tolerance, would discharge the [A] tags that depend on it alone; the soft conditioned-law limit depends on the stronger Ansatz B and its tags remain until that is closed; the numerics above make the outcome unsurprising without making it a theorem. The deep-tail computation with harmonic-function importance sampling would supply the one constant this paper leaves unevaluated. The non-positive-response theorem's exponential rate

is established at frozen background only, and whether an unconditional rate holds is open. And the recurrent-driver proposition and the two-condition theorem are settled for the Brownian driver, the fundamental null-recurrent case, and open for null-recurrent drivers in general.

## Provenance

*On Why two channels.* The asteroid observation behind the two-channel split is the author's, from 2021; the internal/external decomposition of the Departure Point is in print in [18].

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